Math 101 Exam 2 Extra Credit

Show complete handwritten work for all 50 questions on separate paper. Answers are in red; answers alone do not earn credit.

1. Factor completely:

$9x^2 - 16$

Answer:

$(3x - 4)(3x + 4)$

2. Factor completely:

$25x^2 - 49y^2$

Answer:

$(5x - 7y)(5x + 7y)$

3. Factor completely:

$64m^2 - 81n^2$

Answer:

$(8m - 9n)(8m + 9n)$

4. Solve the inequality and write the answer in interval notation:

$-8 \le 2x - 2 \le 6$

Answer:

$[-3, 4]$

5. Solve the inequality and write the answer in interval notation:

$-13 < 3x + 2 \le 11$

Answer:

$(-5, 3]$

6. Solve the inequality and write the answer in interval notation:

$5 \le 4x + 1 < 17$

Answer:

$[1, 4)$

7. Solve and write the solution in interval notation:

$-17x + 5 \le -9(x - 3)$

Answer:

$\left[-\frac{11}{4}, \infty\right)$

8. Solve and write the solution in interval notation:

$4(5x - 3) > 7x + 14$

Answer:

$(2, \infty)$

9. Solve and write the solution in interval notation:

$-6x + 11 \ge 3(x - 4)$

Answer:

$\left(-\infty, \frac{23}{9}\right]$

10. Multiply and simplify completely:

$(x - 4)(x^2 + 2x + 3)$

Answer:

$x^3 - 2x^2 - 5x - 12$

11. Multiply and simplify completely:

$(x + 3)(x^2 - 5x + 2)$

Answer:

$x^3 - 2x^2 - 13x + 6$

12. Multiply and simplify completely:

$(-2h^2 + 3)^2$

Answer:

$4h^4 - 12h^2 + 9$

13. Solve and give your answer as an interval:

$2|x - 3| + 1 \le 9$

Answer:

$[-1, 7]$

14. Solve and give your answer as an interval:

$3|x + 2| - 5 < 7$

Answer:

$(-6, 2)$

15. Solve and give your answer in interval notation:

$|4x - 1| \ge 7$

Answer:

$\left(-\infty, -\frac{3}{2}\right] \cup [2, \infty)$

16. Factor completely:

$x^3 + 64$

Answer:

$(x + 4)(x^2 - 4x + 16)$

17. Factor completely:

$8x^3 - 27$

Answer:

$(2x - 3)(4x^2 + 6x + 9)$

18. Factor completely:

$3x^3 + 375$

Answer:

$3(x + 5)(x^2 - 5x + 25)$

19. Factor by grouping:

$x^2y + 4y + 3x^2 + 12$

Answer:

$(x^2 + 4)(y + 3)$

20. Factor by grouping:

$xy + 6x - 5y - 30$

Answer:

$(x - 5)(y + 6)$

21. Factor by grouping:

$2a^2b + 10b + 3a^2 + 15$

Answer:

$(a^2 + 5)(2b + 3)$

22. Solve and write the answer in interval notation:

$x + \frac{5}{4} < \frac{11}{6}$

Answer:

$\left(-\infty, \frac{7}{12}\right)$

23. Solve and write the answer in interval notation:

$\frac{3x + 2}{-4} \ge 5$

Answer:

$\left(-\infty, -\frac{22}{3}\right]$

24. Solve and write the answer in interval notation:

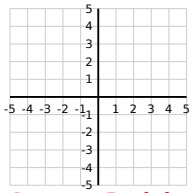
$\frac{2x-1}{3} > \frac{7}{2}$

Answer:

$\left(\frac{23}{4}, \infty\right)$

25. Graph the system:

$y \geq x - 2, \quad y \leq -x + 4$

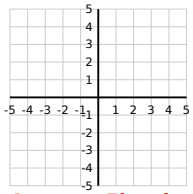


Answer: Both boundaries solid; shade above the first line and below the second.

$(3, 1)$

26. Graph the system:

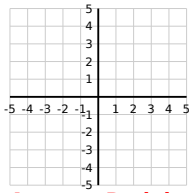
$y > 2x - 1, \quad y \leq -x + 5$



Answer: First boundary dashed; second boundary solid. Shade the overlap.

27. Graph the system:

$-x + y \leq 2, \quad 2x + y \geq -1$



Answer: Both boundaries solid. Shade the overlap.

28. Solve the compound inequality and write the answer in interval notation:

$2x - 5 \leq -9 \quad \text{OR} \quad 3x + 1 > 10$

Answer:

$(-\infty, -2] \cup (3, \infty)$

29. Solve the compound inequality and write the answer in interval notation:

$-4x + 3 \geq 15 \quad \text{OR} \quad 5x - 2 < 8$

Answer:

$(-\infty, 2)$

30. Solve the compound inequality and write the answer in interval notation:

$6x + 1 < -11 \quad \text{OR} \quad -2x + 5 \leq 1$

Answer:

$(-\infty, -2) \cup [2, \infty)$

31. Solve the equation:

$x(x - 7) = 18$

Answer:

$x = -2, \ 9$

32. Solve the equation:

$(m + 8)(m + 2) = 40$

Answer:

$m = -12, \ 2$

33. Solve the equation:

$q^2 - 9q = -20$

Answer:

$q = 4, \ 5$

34. Solve the equation:

$|3x - 2| + 4 = 19$

Answer:

$x = -\frac{13}{3}, \ \frac{17}{3}$

35. Solve the equation:

$2|5x + 1| = 12$

Answer:

$x = -\frac{7}{5}, \ 1$

36. Solve the equation:

$-3 - 2|4x - 5| = -17$

Answer:

$x = -\frac{1}{2}, \ 3$

37. Factor completely:

$6x^2 + 13x + 6$

Answer:

$(3x + 2)(2x + 3)$

38. Factor completely:

$10x^2 - 27x + 5$

Answer:

$(2x - 5)(5x - 1)$

39. Factor completely:

$12x^2 + 11x - 5$

Answer:

$(3x - 1)(4x + 5)$

40. Factor the polynomial completely:

$t^6 + 5t^5 - 6t^4$

Answer:

$t^4(t + 6)(t - 1)$

41. Factor the polynomial completely:

$2x^6 + 14x^5 + 24x^4$

Answer:

$2x^4(x + 3)(x + 4)$

42. Factor the polynomial completely:

$3m^5 - 27m^3$

Answer:

$3m^3(m - 3)(m + 3)$

43. Subtract the polynomials and simplify completely:

$(8x^3 - 2x^2 + 5x - 1) - (3x^4 + 4x^3 - x^2 + 7)$

Answer:

$-3x^4 + 4x^3 - x^2 + 5x - 8$

44. Subtract the polynomials and simplify completely:

$(-2x^5 + 6x^2 - 3x + 4) - (-5x^4 + x^2 + 8x - 9)$

Answer:

$-2x^5 + 5x^4 + 5x^2 - 11x + 13$

45. Subtract the polynomials and simplify completely:

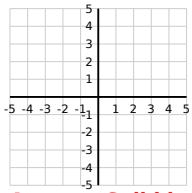
$(7x^4 - 3x^3 + 2) - (2x^4 + 5x^3 - 6x + 1)$

Answer:

$5x^4 - 8x^3 + 6x + 1$

46. Graph:

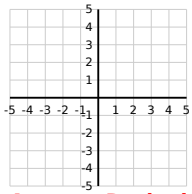
$y \geq \frac{1}{2}x - 2$



Answer: Solid boundary; shade above the line.

47. Graph:

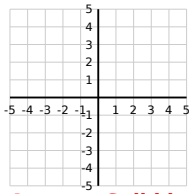
$y < -2x + 3$



Answer: Dashed boundary; shade below the line.

48. Graph:

$2x + y \leq 4$



Answer: Solid boundary; shade below the line.

49. The product of two consecutive page numbers is 306. Find the page numbers:

Answer:

17 and 18

50. The product of two consecutive page numbers is 420. Find the page numbers:

Answer:

20 and 21